

GEARS Human Validation — Reviewer 2 (Blinded)

Purpose

Independently determine how each citing paper uses GEARS. Machine labels, confidence, agent outputs, and machine-selected evidence are intentionally hidden.

Workflow

Complete every row in the displayed order. Do not open the other reviewer document or the machine-selection file until both independent reviews are locked.
Use only the linked approved article versions. Inspect every GEARS occurrence and its surrounding methods, results, figures, tables, and available supplements.
After both reviewers lock their answers, discuss only disagreements in a separate consensus record.

Allowed values

Review status: DRAFT / LOCKED
Citation validity and method executed: Yes / No / Unclear
Uses (select all): Mention only / Benchmark / Extend or develop / Apply to biological data / Other executed use / Unclear
Primary use: Mention only / Benchmark / Extend or develop / Apply to biological data / Other / Unclear
Biological insight: Yes / No / Unclear
Evidence sufficiency: Sufficient / Incomplete / Citation not located / Unclear
Supporting evidence: shortest exact quote plus section, page, paragraph, figure, or table locator.

Review table

Record	Article and approved evidence links	Reviewer 2 independent record
GEARS-HV-005 FORMAL VALIDATION	GexMolGen: cross-modal generation of hit-like molecules via large language model encoding of gene expression signatures DOI: 10.1093/bib/bbae525 Article page: https://doi.org/10.1093/bib/bbae525 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/11/26/2023.11.11.566725.full.pdf No controller-registered target occurrence; independently audit citation location and evidence coverage.	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We further utilize the mask language model (MLM) task to fine-tune scGPT. Instead of commonly using mean square loss, we use the autofocus direction-aware loss in GEARS [32]. Notes:
GEARS-HV-020 FORMAL VALIDATION	Fluctuation structure predicts genome-wide perturbation outcomes DOI: 10.1101/2025.06.27.661814 Article page: https://doi.org/10.1101/2025.06.27.661814 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/07/01/2025.06.27.661814.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_726b71ecff5ac6626a97.txt:p-0007); occ-0002 (document_726b71ecff5ac6626a97.txt:p-0082)	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We benchmarked against five recent approaches spanning distinct methodological classes—GenePert ⁵¹ , scLAMBDA ²¹ , Scouter ⁵² , scGPT ²⁰ , and GEARS ⁵³ —as well as a linear mean baseline, motivated by recent concerns that simple linear models can match far more complex architectures on this task ⁵⁴ . T Notes:

GEARS-HV-043 FORMAL VALIDATION	Predicting Multigene Transcriptional Outcomes Based on Dual-Stream Mamba Neural Network DOI: 10.12677/biphy.2025.131001 Article page: https://doi.org/10.12677/biphy.2025.131001 Full text 1: https://drpress.org/ojs/index.php/fcis/article/download/31264/30614 Full text 2: https://pdf.hanspub.org/biphy_2790105.pdf Controller-registered target occurrence locator(s): occ-0001 (document_6ff398dc26411036850e.txt:p-0004); occ-0002 (document_6ff398dc26411036850e.txt:p-0005); occ-0003 (document_6ff398dc26411036850e.txt:p-0005); occ-0004 (document_6ff398dc26411036850e.txt:p-0008); occ-0005 (document_6ff398dc26411036850e.txt:p-0024); occ-0006 (document_fbe3d2050efb71b5d261.txt:p-0018); occ-0007 (document_fbe3d2050efb71b5d261.txt:p-0018); occ-0008 (document_fbe3d2050efb71b5d261.txt:p-0023); occ-0009 (document_fbe3d2050efb71b5d261.txt:p-0051)	Review status: Citation validity: Yes Method executed: Uses (select all): Primary use: Biological insight: Evidence sufficiency: No Supporting quote + locator: Notes:
GEARS-HV-010 FORMAL VALIDATION	On the use of variational autoencoders for biomedical data integration DOI: 10.1101/2025.08.18.670835 Article page: https://doi.org/10.1101/2025.08.18.670835 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/08/22/2025.08.18.670835.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_136c9cb018fe55dbb9ba.txt:p-0139)	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: we compared MOVE's performance with that of alternative methods including GEARS, CPA, scGPT and their baseline linear models. T Notes:
GEARS-HV-009 FORMAL VALIDATION	Unagi: Deep Generative Model for Deciphering Cellular Dynamics and In-Silico Drug Discovery in Complex Diseases DOI: 10.21203/rs.3.rs-3676579/v1 Article page: https://doi.org/10.21203/rs.3.rs-3676579/v1 Full text 1: https://www.researchsquare.com/article/rs-3676579/latest.pdf Controller-registered target occurrence locator(s): occ-0001 (document_f5992f36e601995af4bb.txt:p-0004); occ-0002 (document_f5992f36e601995af4bb.txt:p-0032)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: nt gap in current single-cell methodologies is the absence of 74 unsupervised in-silico exploration capabilities. Although GEARS 25 and scGen 23 are able to perform in-silico 75

		perturbations, they weren't designed to process time-series data and require the in-vitro screening of 76 perturbation response of disease and drug treatments as the supervision. Thus, there is a pressing need for 77 methods that can virtually examine thousands of potential drugs and compounds on single-cell disease data 78 without ground truth training data. Notes:
GEARS-HV-018 FORMAL VALIDATION	xTrimoGene: An Efficient and Scalable Representation Learner for Single-Cell RNA-Seq Data DOI: 10.1101/2023.03.24.534055 Article page: https://doi.org/10.1101/2023.03.24.534055 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/03/25/2023.03.24.534055.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_5c94df88575250ab20d9.txt:p-0012); occ-0002 (document_5c94df88575250ab20d9.txt:p-0012); occ-0003 (document_5c94df88575250ab20d9.txt:p-0012); occ-0004 (document_5c94df88575250ab20d9.txt:p-0012); occ-0005 (document_5c94df88575250ab20d9.txt:p-0015); occ-0006 (document_5c94df88575250ab20d9.txt:p-0017); occ-0007 (document_5c94df88575250ab20d9.txt:p-0017); occ-0008 (document_5c94df88575250ab20d9.txt:p-0017); occ-0009 (document_5c94df88575250ab20d9.txt:p-0017)	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We compared the native GEARS model with and without incorporating embeddings from xTrimoGene. Notes:
GEARS-HV-044 FORMAL VALIDATION	BIOFORMERS: A SCALABLE FRAMEWORK FOR EXPLORING BIOSTATES USING TRANSFORMERS DOI: 10.1101/2023.11.29.569320 Article page: https://doi.org/10.1101/2023.11.29.569320 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/12/01/2023.11.29.569320.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_c16d75fa50a80feb1a20.txt:p-0013)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: . For this task, our benchmark dataset was the Perturb-seq dataset by Adamson et al. [25], which was preprocessed by Roohani et al. [26]. Notes:
GEARS-HV-040	Medea: An AI agent for therapeutic reasoning across biological contexts DOI: 10.64898/2026.01.16.696667	Review status: Citation validity: No Method executed: Uses (select all):

FORMAL VALIDATION	Article page: https://doi.org/10.64898/2026.01.16.696667 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2026/01/20/2026.01.16.696667.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_22b70643f39817d721ad.txt:p-0036)	Primary use: Biological insight: Evidence sufficiency: No Supporting quote + locator: Notes:
GEARS-HV-038 FORMAL VALIDATION	scYeast: a Biological-knowledge-guided Foundation Model on Yeast Single-Cell Transcriptomics DOI: 10.1101/2025.08.20.671179 Article page: https://doi.org/10.1101/2025.08.20.671179 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/08/25/2025.08.20.671179.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_7216842b65d3d2e24031.txt:p-0025); occ-0002 (document_7216842b65d3d2e24031.txt:p-0067)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Extend or develop Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: we adapted the GEARS [32] framework by using scYeast as the base model for predicting gene perturbation responses (Methods) (Fig. 4a). Notes:
GEARS-HV-035 FORMAL VALIDATION	Benchmarking foundation cell models for post-perturbation RNA-seq prediction DOI: 10.1186/s12864-025-11600-2 Article page: https://doi.org/10.1186/s12864-025-11600-2 Full text 1: https://bmcbgenomics.biomedcentral.com/counter/pdf/10.1186/s12864-025-11600-2 Controller-registered target occurrence locator(s): occ-0001 (document_d61b46782820f1f5f549.txt:p-0002); occ-0002 (document_d61b46782820f1f5f549.txt:p-0010); occ-0003 (document_d61b46782820f1f5f549.txt:p-0010); occ-0004 (document_d61b46782820f1f5f549.txt:p-0010); occ-0005 (document_d61b46782820f1f5f549.txt:p-0010); occ-0006 (document_d61b46782820f1f5f549.txt:p-0010); occ-0007 (document_d61b46782820f1f5f549.txt:p-0010); occ-0008 (document_d61b46782820f1f5f549.txt:p-0013)	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We reproduced scGPT's results on the GEARS benchmarking datasets using the cell-gears v0.0.1 package and a fork of scGPT v0.2.1 (commit 7301b51). Notes:
GEARS-HV-036 FORMAL VALIDATION	World Models for Pharmaceutical Science Should Simulate Biological Consequences rather than Merely Continue Statistical Patterns DOI: 10.51847/efvx6kqqna Article page: https://doi.org/10.51847/efvx6kqqna Full text 1: https://pharmacophorejournal.com/storage/files/article/5bd92b3d-cf40-41bc-80a2-5a1214db27a5-fP5jJbFj6Ln5v3zj/iZJfMPXRrTmfSTg.pdf Controller-registered target occurrence locator(s): occ-0001 (document_fc4ac8fd9e16858adcd0.txt:p-0027)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Genetic interaction screens provide rich single-cell phenotypes for testing whether a model recovers nonlinear interaction structure; multiplex chemical transcriptomics measures heterogeneous cellular responses to pharmacological perturbations; and prediction of unseen multigene interventions tests whether a model generalizes beyond conditions explicitly represented during training [23-25].

		Notes:
GEARS-HV-034 FORMAL VALIDATION	Knowledge graph-aided Bayesian active learning for top-K genetic interaction discovery DOI: 10.1038/s41598-025-13972-7 Article page: https://doi.org/10.1038/s41598-025-13972-7 Full text 1: https://www.nature.com/articles/s41598-025-13972-7.pdf Controller-registered target occurrence locator(s): occ-0001 (document_01075fd12409c5a74d4f.txt:p-0033)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: For gene knockdown experiments, active learning strategies have been investigated in several studies ^{3,4,5,6,7,8} . Notes:
GEARS-HV-046 FORMAL VALIDATION	Generative genomics accurately predicts future experimental results DOI: 10.1101/2025.09.08.674753 Article page: https://doi.org/10.1101/2025.09.08.674753 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/09/09/2025.09.08.674753.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_982567858905463cf522.txt:p-0093)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Fourth, while our models encompass a wide array of gene expression experiments, we did not include extensive Perturb-seq data during training. Despite being a valuable resource for AI model development ^{20,34,35} , Perturb-seq data presents unique challenges, including technical artifacts and non-specific responses ^{20,36–39} , which require focused resolution for this data type that goes beyond the scope of our initial study. Notes:
GEARS-HV-004 FORMAL VALIDATION	Multimodal Benchmarking of Foundation Model Representations for Cellular Perturbation Response Prediction DOI: 10.1101/2025.06.26.661186 Article page: https://doi.org/10.1101/2025.06.26.661186 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/06/28/2025.06.26.661186.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_4b239dec33d56e05c63a.txt:p-0006)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: There are many specialized perturbation response models one could use on top of such embeddings (e.g. Roohani et al. (2024), Lotfollahi et al. (2023), Piran et al. (2024)), but those models are beyond the scope of our work, which intentionally fixes a single model with a consistent training protocol. Notes:
GEARS-HV-026	HyperdCas12a-based multiplexed genetic regulation in Candida albicans DOI: 10.1093/nar/gkaf1402	Review status: Citation validity: Yes Method executed: No

FORMAL VALIDATION	Article page: https://doi.org/10.1093/nar/gkaf1402 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/10/20/2025.10.20.683421.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_5d55b450a4b894ee372a.txt:p-0094)	Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Recent technological advances in model systems have enabled the identification of a myriad of genetic interactions and non-additive phenotypic outcomes caused by changes in two or more genes [17–21]. Notes:
GEARS-HV-033 FORMAL VALIDATION	Molecular causality in the advent of foundation models DOI: 10.1038/s44320-024-00041-w Article page: https://doi.org/10.1038/s44320-024-00041-w Full text 1: https://link.springer.com/content/pdf/10.1038/s44320-024-00041-w.pdf Controller-registered target occurrence locator(s): occ-0001 (document_43a44b24877ae33f7710.txt:p-0010)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Hybrid models make a tradeoff between those extremes, which is why they have been found to be useful in systems biology, where data are currently scarce (AlQuraishi and Sorger, 2021; Nilsson et al, 2022; Faure et al, 2023; Roohani et al, 2023; Notes:
GEARS-HV-008 FORMAL VALIDATION	scFormer: A Universal Representation Learning Approach for Single-Cell Data Using Transformers DOI: 10.1101/2022.11.20.517285 Article page: https://doi.org/10.1101/2022.11.20.517285 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2022/11/22/2022.11.20.517285.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_c8dd8f4837f2553f5226.txt:p-0009); occ-0002 (document_c8dd8f4837f2553f5226.txt:p-0009); occ-0003 (document_c8dd8f4837f2553f5226.txt:p-0011)	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We compare the performance against the recent GEARS method (Roohani et al., 2022) and the multi layer perceptron baseline. Notes:
GEARS-HV-019 FORMAL VALIDATION	Knowledge-augmented Graph Machine Learning for Drug Discovery: A Survey DOI: 10.1145/3744237 Article page: https://doi.org/10.1145/3744237 Full text 1: https://pure.au.dk/ws/files/469052743/Knowledge-augmented_Graph_Machine_Learning_for_Drug_Discovery.pdf Controller-registered target occurrence locator(s): occ-0001 (document_ba89fde9a7c4cd903d44.txt:p-0015);	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Mentioned in table Notes:

	occ-0002 (document_ba89fde9a7c4cd903d44.txt:p-0034)	
GEARS-HV-013 FORMAL VALIDATION	A systematic comparison of computational methods for expression forecasting DOI: 10.1101/2023.07.28.551039 Article page: https://doi.org/10.1101/2023.07.28.551039 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/07/31/2023.07.28.551039.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_02100be951234bc62ab3.txt:p-0005); occ-0002 (document_02100be951234bc62ab3.txt:p-0006); occ-0003 (document_02100be951234bc62ab3.txt:p-0018); occ-0004 (document_02100be951234bc62ab3.txt:p-0020); occ-0005 (document_02100be951234bc62ab3.txt:p-0020); occ-0006 (document_02100be951234bc62ab3.txt:p-0021); occ-0007 (document_02100be951234bc62ab3.txt:p-0021); occ-0008 (document_02100be951234bc62ab3.txt:p-0021); occ-0009 (document_02100be951234bc62ab3.txt:p-0021); occ-0010 (document_02100be951234bc62ab3.txt:p-0021); occ-0011 (document_02100be951234bc62ab3.txt:p-0021); occ-0012 (document_02100be951234bc62ab3.txt:p-0021); occ-0013 (document_02100be951234bc62ab3.txt:p-0021); occ-0014 (document_02100be951234bc62ab3.txt:p-0021); occ-0015 (document_02100be951234bc62ab3.txt:p-0022); occ-0016 (document_02100be951234bc62ab3.txt:p-0022); occ-0017 (document_02100be951234bc62ab3.txt:p-0022); occ-0018 (document_02100be951234bc62ab3.txt:p-0023); occ-0019 (document_02100be951234bc62ab3.txt:p-0023); occ-0020 (document_02100be951234bc62ab3.txt:p-0024); occ-0021 (document_02100be951234bc62ab3.txt:p-0025); occ-0022 (document_02100be951234bc62ab3.txt:p-0025); occ-0023 (document_02100be951234bc62ab3.txt:p-0027); occ-0024 (document_02100be951234bc62ab3.txt:p-0027); occ-0025 (document_02100be951234bc62ab3.txt:p-0027); occ-0026 (document_02100be951234bc62ab3.txt:p-0027);	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: For adamson, dixit, and norman, we eschew these filters and instead use preprocessing identical to the GEARs evaluations (Roohani et al., 2024). Notes:

	occ-0027 (document_02100be951234bc62ab3.txt:p-0033); occ-0028 (document_02100be951234bc62ab3.txt:p-0039); occ-0029 (document_02100be951234bc62ab3.txt:p-0039); occ-0030 (document_02100be951234bc62ab3.txt:p-0039); occ-0031 (document_02100be951234bc62ab3.txt:p-0039)	
GEARS-HV-028 FORMAL VALIDATION	Causal models and prediction in cell line perturbation experiments DOI: 10.1186/s12859-024-06027-7 Article page: https://doi.org/10.1186/s12859-024-06027-7 Full text 1: https://www.researchsquare.com/article/rs-3973384/latest.pdf Full text 2: https://bmcbioinformatics.biomedcentral.com/content/pdf/10.1186/s12859-024-06027-7 Controller-registered target occurrence locator(s): occ-0001 (document_8e546d952becc9f38df6.txt:p-0018); occ-0002 (document_d18339b76d624d106839.txt:p-0005); occ-0003 (document_d18339b76d624d106839.txt:p-0023)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Several works have developed models for predicting responses to previously tested perturbations in new settings. For example, [7, 8], and [13] developed deep-learning autoencoder models to predict responses to previously untested perturbation-cell type combinations. [6] considered a similar problem and developed prediction model, SI-A, derived from the synthetic control literature in causal inference. [10] developed GEARS to predict the effect of double knockouts/knockdowns using <i>in vitro</i> responses to single knockouts/knockdowns. Notes:
GEARS-HV-031 FORMAL VALIDATION	Simple Causal Relationships in Gene Expression Discovered through Deep Learned Collective Variables DOI: 10.1101/2023.01.18.524617 Article page: https://doi.org/10.1101/2023.01.18.524617 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/01/20/2023.01.18.524617.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_c4e884dd4ef14a20ead1.txt:p-0060)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: The use of autoencoders [9], UMAP [10,11], and other methods of dimensionality reduction [12,13] have demonstrated that it is possible to discover relationships in statistical covariation of many genes that can be used, for example, to distinguish different cell types and lineages [14], to predict mean gene expression under perturbation [15,16], to predict the interactions between genes under combinatorial perturbations [17], and to generate counterfactual

		predictions on single-cell response under new drug treatments [18] or novel perturbations [19]. Notes:
GEARS-HV-048 FORMAL VALIDATION	CellPLM: Pre-training of Cell Language Model Beyond Single Cells DOI: 10.1101/2023.10.03.560734 Article page: https://doi.org/10.1101/2023.10.03.560734 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/10/05/2023.10.03.560734.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_fd1cf2273fcbecd90247.txt:p-0016); occ-0002 (document_fd1cf2273fcbecd90247.txt:p-0025); occ-0003 (document_fd1cf2273fcbecd90247.txt:p-0025); occ-0004 (document_fd1cf2273fcbecd90247.txt:p-0025); occ-0005 (document_fd1cf2273fcbecd90247.txt:p-0025); occ-0006 (document_fd1cf2273fcbecd90247.txt:p-0025); occ-0007 (document_fd1cf2273fcbecd90247.txt:p-0025); occ-0008 (document_fd1cf2273fcbecd90247.txt:p-0026); occ-0009 (document_fd1cf2273fcbecd90247.txt:p-0028)	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We compare the performance between <i>CellPLM</i> and two baselines, i.e., a recent preprint GEARS method (Roohani et al., 2022), and scGen (Lotfollahi et al., 2019). Notes:
GEARS-HV-029 FORMAL VALIDATION	HisToSpatialCNV: an interpretable deep learning method predicting spatial copy number variations from histopathology images DOI: 10.1038/s41551-026-01754-z Article page: https://doi.org/10.1038/s41551-026-01754-z Full text 1: https://www.nature.com/articles/s41551-026-01754-z.pdf Controller-registered target occurrence locator(s): occ-0001 (document_1f68275d2a59b48d816a.txt:p-0018)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Similar observations were also shared by others, in that single-cell FMs currently do not necessarily outperform established task-specific methods across all downstream tasks^{74,75,76,77}. Notes:
GEARS-HV-011 FORMAL VALIDATION	scEMB: Learning context representation of genes based on large-scale single-cell transcriptomics DOI: 10.1101/2024.09.24.614685 Article page: https://doi.org/10.1101/2024.09.24.614685 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/09/26/2024.09.24.614685.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_084a1ab5ec97944eb131.txt:p-0020)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Recent advances in artificial intelligence (AI), such as variational autoencoder (VAE) models^{8–10}, which explicitly learn low-dimensional embeddings from single-cell transcriptome to capture the biological

		meaning embedding. Moreover, these embedding-based methods¹¹ [Autoencoder¹², VAE^{13–16}, GNN¹⁷], Notes:
GEARS-HV-049 FORMAL VALIDATION	Decoding cell identity with multi-scale explainable deep learning DOI: 10.1101/2024.02.05.578922 Article page: https://doi.org/10.1101/2024.02.05.578922 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/02/08/2024.02.05.578922.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_c52b63babff855635323.txt:p-0054)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Due to the incorporation of gene/protein interaction information, Cell Decoder opens up the possibility of predicting cell responses to genetic perturbations⁵⁷. Notes:
GEARS-HV-003 FORMAL VALIDATION	rbio1 - training scientific reasoning LLMs with biological world models as soft verifiers DOI: 10.1101/2025.08.18.670981 Article page: https://doi.org/10.1101/2025.08.18.670981 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/08/21/2025.08.18.670981.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_de87e74d675a63fc9796.txt:p-0013); occ-0002 (document_de87e74d675a63fc9796.txt:p-0014); occ-0003 (document_de87e74d675a63fc9796.txt:p-0014); occ-0004 (document_de87e74d675a63fc9796.txt:p-0026)	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We compare against: SUMMER [32] (retrieval + knowledge-based reasoning), GEARS [38] (a specialized perturbation model), and the base Qwen2.5-3B reasoning model without biological supervision. Notes:
GEARS-HV-023 FORMAL VALIDATION	GPerturb: Gaussian process modelling of single-cell perturbation data DOI: 10.1038/s41467-025-61165-7 Article page: https://doi.org/10.1038/s41467-025-61165-7 Full text 1: https://www.nature.com/articles/s41467-025-61165-7.pdf Controller-registered target occurrence locator(s): occ-0001 (document_650988dd5720f43d70f3.txt:p-0002); occ-0002 (document_650988dd5720f43d70f3.txt:p-0002); occ-0003 (document_650988dd5720f43d70f3.txt:p-0002); occ-0004 (document_650988dd5720f43d70f3.txt:p-0002); occ-0005 (document_650988dd5720f43d70f3.txt:p-0002); occ-0006 (document_650988dd5720f43d70f3.txt:p-0002); occ-0007 (document_650988dd5720f43d70f3.txt:p-0003); occ-0008 (document_650988dd5720f43d70f3.txt:p-0003);	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We first compared the predictive performance of GPerturb, CPA, GEARS and SAMS-VAE on a subset of the genome-wide CRISPR interference Perturb-seq dataset.³³ Notes:

	occ-0009 (document_650988dd5720f43d70f3.txt:p-0003); occ-0010 (document_650988dd5720f43d70f3.txt:p-0004); occ-0011 (document_650988dd5720f43d70f3.txt:p-0004); occ-0012 (document_650988dd5720f43d70f3.txt:p-0004); occ-0013 (document_650988dd5720f43d70f3.txt:p-0004); occ-0014 (document_650988dd5720f43d70f3.txt:p-0004); occ-0015 (document_650988dd5720f43d70f3.txt:p-0004); occ-0016 (document_650988dd5720f43d70f3.txt:p-0004); occ-0017 (document_650988dd5720f43d70f3.txt:p-0004); occ-0018 (document_650988dd5720f43d70f3.txt:p-0004); occ-0019 (document_650988dd5720f43d70f3.txt:p-0004); occ-0020 (document_650988dd5720f43d70f3.txt:p-0004); occ-0021 (document_650988dd5720f43d70f3.txt:p-0004); occ-0022 (document_650988dd5720f43d70f3.txt:p-0005); occ-0023 (document_650988dd5720f43d70f3.txt:p-0005); occ-0024 (document_650988dd5720f43d70f3.txt:p-0005); occ-0025 (document_650988dd5720f43d70f3.txt:p-0005); occ-0026 (document_650988dd5720f43d70f3.txt:p-0005); occ-0027 (document_650988dd5720f43d70f3.txt:p-0005); occ-0028 (document_650988dd5720f43d70f3.txt:p-0005); occ-0029 (document_650988dd5720f43d70f3.txt:p-0007); occ-0030 (document_650988dd5720f43d70f3.txt:p-0007); occ-0031 (document_650988dd5720f43d70f3.txt:p-0012); occ-0032 (document_650988dd5720f43d70f3.txt:p-0012); occ-0033 (document_650988dd5720f43d70f3.txt:p-0012); occ-0034 (document_650988dd5720f43d70f3.txt:p-0012); occ-0035 (document_650988dd5720f43d70f3.txt:p-0013)	
GEARS-HV-014 FORMAL VALIDATION	scMomer: A modality-aware pretraining framework for single-cell multi-omics modeling under missing modality conditions DOI: 10.1101/2025.08.04.668374 Article page: https://doi.org/10.1101/2025.08.04.668374 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/08/05/2025.08.04.668374.full.pdf	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Results showed that scMomer outperformed GEARS, achieving

	Controller-registered target occurrence locator(s): occ-0001 (document_32bff1ac12fad55c6696.txt:p-0018); occ-0002 (document_32bff1ac12fad55c6696.txt:p-0018); occ-0003 (document_32bff1ac12fad55c6696.txt:p-0025); occ-0004 (document_32bff1ac12fad55c6696.txt:p-0026); occ-0005 (document_32bff1ac12fad55c6696.txt:p-0026); occ-0006 (document_32bff1ac12fad55c6696.txt:p-0029)	improvements of 0.034 in PCC (Supplementary Table 5). Notes:
GEARS-HV-024 FORMAL VALIDATION	scPRINT: pre-training on 50 million cells allows robust gene network predictions DOI: 10.1038/s41467-025-58699-1 Article page: https://doi.org/10.1038/s41467-025-58699-1 Full text 1: https://www.nature.com/articles/s41467-025-58699-1.pdf Controller-registered target occurrence locator(s): occ-0001 (document_59b9ce377d6f923515c3.txt:p-0020)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Understanding the cellular mechanism is considered a milestone in biology, allowing us to predict cell behavior and the impact of drugs and gene knock-outs^{1,2,3,4,5}. Notes:
GEARS-HV-007 FORMAL VALIDATION	Impact of the Human Cell Atlas on medicine DOI: 10.1038/s41591-022-02104-7 Article page: https://doi.org/10.1038/s41591-022-02104-7 Full text 1: https://www.nature.com/articles/s41591-022-02104-7.pdf Controller-registered target occurrence locator(s): occ-0001 (document_80ad0abf109b97c993e8.txt:p-1226)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Moreover, machine-learning algorithms can increasingly be trained on such data to yield models that predict the impact of additional perturbations in one or more genes in the same cellular context or of the same perturbations in new biological contexts^{114,115,116}. Notes:
GEARS-HV-025 FORMAL VALIDATION	Mapping ovarian cancer spatial organization uncovers immune evasion drivers at the genetic, cellular, and tissue level DOI: 10.1101/2023.10.16.562592 Article page: https://doi.org/10.1101/2023.10.16.562592 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/10/19/2023.10.16.562592.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_d96dc927725a902640fe.txt:p-0115)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: it will be possible to use Perturb-seq data more effectively to extrapolate from one context to another with increasing accuracy^{70,71} Notes:

GEARS-HV-021 FORMAL VALIDAT ION	Multifaceted Representation of Genes via Deep Learning of Gene Expression Networks DOI: 10.1101/2024.03.07.583777 Article page: https://doi.org/10.1101/2024.03.07.583777 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/03/09/2024.03.07.583777.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_ce413a5d2d0e6225b875.txt:p-0013)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Accurate predictive modeling of human gene relationships would fundamentally transform our ability to uncover the molecular mechanisms that underpin key biological and disease processes. Recent studies have employed advanced AI techniques to model the complexities of gene networks using large gene expression datasets ^{1–11} . Notes:
GEARS-HV-016 FORMAL VALIDAT ION	Systema: a framework for evaluating genetic perturbation response prediction beyond systematic variation DOI: 10.1038/s41587-025-02777-8 Article page: https://doi.org/10.1038/s41587-025-02777-8 Full text 1: https://www.nature.com/articles/s41587-025-02777-8.pdf Controller-registered target occurrence locator(s): occ-0001 (document_7f2e411f81d7a4258a23.txt:p-0003); occ-0002 (document_7f2e411f81d7a4258a23.txt:p-0003); occ-0003 (document_7f2e411f81d7a4258a23.txt:p-0004); occ-0004 (document_7f2e411f81d7a4258a23.txt:p-0004); occ-0005 (document_7f2e411f81d7a4258a23.txt:p-0007); occ-0006 (document_7f2e411f81d7a4258a23.txt:p-0007); occ-0007 (document_7f2e411f81d7a4258a23.txt:p-0007); occ-0008 (document_7f2e411f81d7a4258a23.txt:p-0008); occ-0009 (document_7f2e411f81d7a4258a23.txt:p-0008); occ-0010 (document_7f2e411f81d7a4258a23.txt:p-0008); occ-0011 (document_7f2e411f81d7a4258a23.txt:p-0009); occ-0012 (document_7f2e411f81d7a4258a23.txt:p-0010); occ-0013 (document_7f2e411f81d7a4258a23.txt:p-0012); occ-0014 (document_7f2e411f81d7a4258a23.txt:p-0012); occ-0015 (document_7f2e411f81d7a4258a23.txt:p-0012); occ-0016 (document_7f2e411f81d7a4258a23.txt:p-0012); occ-0017	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We next evaluated performance on the task of predicting transcriptional responses to two-gene perturbations in the Norman ¹⁴ dataset, the only dataset with combinatorial perturbations in our benchmark. For unseen two-gene perturbations (two-gene perturbations where none of the individual perturbations was observed at train time), the matching mean baseline outperformed all other baselines by a considerable margin, with relative improvements of 11% for Pearson _d over the best alternative method (GEARS; Fig. 1d). Notes:

	(document_7f2e411f81d7a4258a23.txt:p-0013); occ-0018 (document_7f2e411f81d7a4258a23.txt:p-0013); occ-0019 (document_7f2e411f81d7a4258a23.txt:p-0013); occ-0020 (document_7f2e411f81d7a4258a23.txt:p-0013); occ-0021 (document_7f2e411f81d7a4258a23.txt:p-0014)	
GEARS-HV-027 FORMAL VALIDATION	Prediction of cellular morphology change under perturbations with transcriptome-guided diffusion model DOI: 10.1101/2025.07.16.665106 Article page: https://doi.org/10.1101/2025.07.16.665106 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/07/16/2025.07.16.665106.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_4606fd53ac6e26c3d47a.txt:p-0021); occ-0002 (document_4606fd53ac6e26c3d47a.txt:p-0031)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Therefore, integrating MorphDiff with methods that can predict changes in gene expression under unseen perturbations, such as GEARs⁴⁷, is a possible extension. Notes:
GEARS-HV-037 FORMAL VALIDATION	Perturbomics: CRISPR–Cas screening-based functional genomics approach for drug target discovery DOI: 10.1038/s12276-025-01487-0 Article page: https://doi.org/10.1038/s12276-025-01487-0 Full text 1: https://www.nature.com/articles/s12276-025-01487-0.pdf Controller-registered target occurrence locator(s): occ-0001 (document_77500ed58c4fa6921e2d.txt:p-0010); occ-0002 (document_77500ed58c4fa6921e2d.txt:p-0010); occ-0003 (document_77500ed58c4fa6921e2d.txt:p-0012)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Leskovec and colleagues, for instance, developed the graph-enhanced gene activation and repression simulator (GEARS), which integrates CRISPRa screening data with previously published Perturb-seq datasets to predict combinatorial perturbation effects on gene interactions¹⁰⁰. Notes:
GEARS-HV-041 FORMAL VALIDATION	The manatee variational autoencoder model for predicting gene expression alterations caused by transcription factor perturbations DOI: 10.1038/s41598-024-62620-z Article page: https://doi.org/10.1038/s41598-024-62620-z Full text 1: https://www.nature.com/articles/s41598-024-62620-z.pdf Controller-registered target occurrence locator(s): occ-0001 (document_3f8615fddb8ee5b226d1.txt:p-0001); occ-0002 (document_3f8615fddb8ee5b226d1.txt:p-0006)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: CPA³ autoencoder, together with GEARs⁴ and graphVCI⁵ graph neural network models disentangle perturbation covariates with basal cellular states, thus

		providing more flexibility for the in silico perturbation analysis. Notes:
GEARS-HV-047 FORMAL VALIDATION	Prediction of cellular morphology changes under perturbations with a transcriptome-guided diffusion model DOI: 10.1038/s41467-025-63478-z Article page: https://doi.org/10.1038/s41467-025-63478-z Full text 1: https://www.nature.com/articles/s41467-025-63478-z.pdf Controller-registered target occurrence locator(s): occ-0001 (document_39a6578d531a744400b1.txt:p-0011); occ-0002 (document_39a6578d531a744400b1.txt:p-0017)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Therefore, integrating MorphDiff with methods that can predict changes in gene expression under unseen perturbations, such as GEARs⁴⁷, is a possible extension. S Notes:
GEARS-HV-022 FORMAL VALIDATION	Learning Cross-Domain Representations for Transferable Drug Perturbations on Single-Cell Transcriptional Responses DOI: 10.1609/aaai.v39i18.34073 Article page: https://doi.org/10.1609/aaai.v39i18.34073 Full text 1: https://ojs.aaai.org/index.php/AAAI/article/download/34073/36228 Controller-registered target occurrence locator(s): occ-0001 (document_5ee8066e213823971bab.txt:p-0002); occ-0002 (document_5ee8066e213823971bab.txt:p-0009); occ-0003 (document_5ee8066e213823971bab.txt:p-0010); occ-0004 (document_5ee8066e213823971bab.txt:p-0010); occ-0005 (document_5ee8066e213823971bab.txt:p-0010); occ-0006 (document_5ee8066e213823971bab.txt:p-0010); occ-0007 (document_5ee8066e213823971bab.txt:p-0012)	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Notably, these dual-gene perturbations were also chosen for testing in the GEARs study (Roohani, Huang, and Leskovec 2022), thus justifying a robust and objective performance comparison. We trained our model on the expr Notes:
GEARS-HV-050 FORMAL VALIDATION	Perturbation Atlases as Pharmacological Reasoning Engines for Connecting Molecular Interventions to Context-Dependent Cellular Responses DOI: 10.51847/a1xh4xuw5o Article page: https://doi.org/10.51847/a1xh4xuw5o Full text 1: https://pharmacophorejournal.com/storage/files/article/aa68d254-a799-4a9c-8a30-0f7cf119a88a-Y08w047JPZKDeQ4r/9rBBC2PqJpy90ji.pdf Controller-registered target occurrence locator(s): occ-0001 (document_b1433a11e3f2f5132e34.txt:p-0015); occ-0002 (document_b1433a11e3f2f5132e34.txt:p-0022)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: GEARS shows how graph-based information can contribute to prediction of transcriptional outcomes for novel multigene perturbations [16]. Notes:

GEARS-HV-039 FORMAL VALIDATION	Cell Decoder: decoding cell identity with multi-scale explainable deep learning DOI: 10.1186/s13059-025-03832-y Article page: https://doi.org/10.1186/s13059-025-03832-y Full text 1: https://genomebiology.biomedcentral.com/counter/pdf/10.1186/s13059-025-03832-y Controller-registered target occurrence locator(s): occ-0001 (document_86b5e3affed65eab298.txt:p-0029)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Cell Decoder opens up the possibility of predicting cell responses to genetic perturbations [42]. Notes:
GEARS-HV-001 FORMAL VALIDATION	Modeling interpretable correspondence between cell state and perturbation response with CellCap DOI: 10.1101/2024.03.14.585078 Article page: https://doi.org/10.1101/2024.03.14.585078 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/03/16/2024.03.14.585078.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_445c5329884eb9425d2f.txt:p-0003); occ-0002 (document_445c5329884eb9425d2f.txt:p-0003); occ-0003 (document_445c5329884eb9425d2f.txt:p-0011); occ-0004 (document_445c5329884eb9425d2f.txt:p-0015); occ-0005 (document_445c5329884eb9425d2f.txt:p-0015); occ-0006 (document_445c5329884eb9425d2f.txt:p-0024)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: For example, scGen uses a deep generative model to predict the impact of one perturbation on a new cell population [10], and GEARS takes advantage of prior biological knowledge to model nonlinear gene interactions and nonlinearities in response to multiple perturbations [11] Notes:
GEARS-HV-017 FORMAL VALIDATION	A General Single-Cell Analysis Framework via Conditional Diffusion Generative Models DOI: 10.1101/2023.10.13.562243 Article page: https://doi.org/10.1101/2023.10.13.562243 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/10/17/2023.10.13.562243.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_fa1b8e373828243f0f02.txt:p-0001); occ-0002 (document_fa1b8e373828243f0f02.txt:p-0013); occ-0003 (document_fa1b8e373828243f0f02.txt:p-0013); occ-0004 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0005 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0006 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0007 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0008 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0009	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We used the official GEARS code base ³ to reproduce the performance on the three benchmarking datasets. We need to rerun the experiments because the data splits used from the original paper were not published. Thus, directly comparing our results with the reported metrics from the GEARS paper is infeasible. Notes:

	(document_fa1b8e373828243f0f02.txt:p-0018); occ-0010 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0011 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0012 (document_fa1b8e373828243f0f02.txt:p-0022); occ-0013 (document_fa1b8e373828243f0f02.txt:p-0039); occ-0014 (document_fa1b8e373828243f0f02.txt:p-0039); occ-0015 (document_fa1b8e373828243f0f02.txt:p-0039); occ-0016 (document_fa1b8e373828243f0f02.txt:p-0039)	
GEARS-HV-045 FORMAL VALIDATION	Primer on machine learning applications in brain immunology DOI: 10.3389/fbinf.2025.1554010 Article page: https://doi.org/10.3389/fbinf.2025.1554010 Full text 1: https://www.frontiersin.org/journals/bioinformatics/articles/10.3389/fbinf.2025.1554010/pdf Controller-registered target occurrence locator(s): occ-0001 (document_ec05b2c2a5b977b9f169.txt:p-0007)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: For example, machine learning tools can be used to enhance dimensionality reduction and integration of large datasets, but also for more sophisticated tasks, such as predictive modeling of gene perturbations (Xu et al., 2021; Lotfollahi et al., 2022; Roohani et al., 2024). Notes:
GEARS-HV-032 FORMAL VALIDATION	Predicting transcriptional responses to novel chemical perturbations using deep generative model DOI: 10.21203/rs.3.rs-3917469/v1 Article page: https://doi.org/10.21203/rs.3.rs-3917469/v1 Full text 1: https://www.researchsquare.com/article/rs-3917469/latest.pdf Controller-registered target occurrence locator(s): occ-0001 (document_dfc4c6b86b40bc6e8ffb.txt:p-0004); occ-0002 (document_dfc4c6b86b40bc6e8ffb.txt:p-0025)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: r chemical perturbations of diverse cell 63 type and compound combinations. GEARS and CellOracle [11 , 12] leveraged 64 a knowledge graph of gene-gene relationships to predict genetic perturbation 65 outcomes. Notes:

GEARS-HV-030 FORMAL VALIDATION	Imputing spatial transcriptomics through gene network constructed from protein language model DOI: 10.1038/s42003-024-06964-2 Article page: https://doi.org/10.1038/s42003-024-06964-2 Full text 1: https://www.nature.com/articles/s42003-024-06964-2.pdf Controller-registered target occurrence locator(s): occ-0001 (document_3d5b0c1d4207d32a4481.txt:p-0011)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Most importantly, TransImpLR ignores gene-to-gene relationships in gene prediction, which have shown improved performance in previous work ^{15,16} . Notes:
GEARS-HV-012 FORMAL VALIDATION	Evaluating the Utilities of Large Language Models in Single-cell Data Analysis DOI: 10.21203/rs.3.rs-3376641/v1 Article page: https://doi.org/10.21203/rs.3.rs-3376641/v1 Full text 1: https://www.researchsquare.com/article/rs-3376641/latest.pdf Controller-registered target occurrence locator(s): occ-0001 (document_94ec69cb17831b26e059.txt:p-0009); occ-0002 (document_94ec69cb17831b26e059.txt:p-0010); occ-0003 (document_94ec69cb17831b26e059.txt:p-0010); occ-0004 (document_94ec69cb17831b26e059.txt:p-0011); occ-0005 (document_94ec69cb17831b26e059.txt:p-0024); occ-0006 (document_94ec69cb17831b26e059.txt:p-0024); occ-0007 (document_94ec69cb17831b26e059.txt:p-0031)	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We considered scGPT and GEARS [42] for this task. In the training process, we masked the genes under perturbation and tried to reconstruct the expression levels of all the genes of the input cells rather than only the masked genes. We used Mean Pearson Correlation (MPC) as the metric to evaluate the performance of scGPT under different hyper-parameters or initial settings, and details can be found in Appendix C. T Notes:
GEARS-HV-015 FORMAL VALIDATION	MultiPert: An adversarial alignment and dual attention framework for single-cell multi-omics perturbation prediction DOI: 10.1371/journal.pcbi.1014054 Article page: https://doi.org/10.1371/journal.pcbi.1014054 Full text 1: https://journals.plos.org/ploscompbiol/article/file?id=10.1371/journal.pcbi.1014054&type=printable Controller-registered target occurrence locator(s): occ-0001 (document_9719522b17e400f49ce6.txt:p-0003); occ-0002 (document_9719522b17e400f49ce6.txt:p-0004); occ-0003 (document_9719522b17e400f49ce6.txt:p-0010); occ-0004 (document_9719522b17e400f49ce6.txt:p-0020);	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Extend or develop Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: . Given GEARS' prior validation of GO knowledge graph for perturbation prediction, we directly adopted their pre-trained embeddings to initialize MultiPert's perturbation representations [16].

	occ-0005 (document_9719522b17e400f49ce6.txt:p-0020); occ-0006 (document_9719522b17e400f49ce6.txt:p-0020); occ-0007 (document_9719522b17e400f49ce6.txt:p-0020); occ-0008 (document_9719522b17e400f49ce6.txt:p-0020); occ-0009 (document_9719522b17e400f49ce6.txt:p-0021); occ-0010 (document_9719522b17e400f49ce6.txt:p-0021); occ-0011 (document_9719522b17e400f49ce6.txt:p-0023)	Notes:
GEARS-HV-042 FORMAL VALIDATION	Harnessing the power of single-cell large language models with parameter-efficient fine-tuning using scPEFT DOI: 10.1038/s42256-025-01170-z Article page: https://doi.org/10.1038/s42256-025-01170-z Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/04/23/2025.04.21.649754.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_186c573a618178144ab2.txt:p-0030); occ-0002 (document_186c573a618178144ab2.txt:p-0030); occ-0003 (document_186c573a618178144ab2.txt:p-0034); occ-0004 (document_186c573a618178144ab2.txt:p-0066); occ-0005 (document_186c573a618178144ab2.txt:p-0066); occ-0006 (document_186c573a618178144ab2.txt:p-0070); occ-0007 (document_186c573a618178144ab2.txt:p-0070); occ-0008 (document_186c573a618178144ab2.txt:p-0070); occ-0009 (document_186c573a618178144ab2.txt:p-0070); occ-0010 (document_186c573a618178144ab2.txt:p-0072); occ-0011 (document_186c573a618178144ab2.txt:p-0072); occ-0012 (document_186c573a618178144ab2.txt:p-0072); occ-0013 (document_186c573a618178144ab2.txt:p-0072); occ-0014 (document_186c573a618178144ab2.txt:p-0100); occ-0015 (document_186c573a618178144ab2.txt:p-0100)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: For perturbation prediction using scFoundation and scGPT, scPEFT enabled scFoundation to overcome GPU memory limitations encountered during fine-tuning. It improved the mean squared error (MSE) and Pearson’s correlation coefficient (PCC) for the top 20 most differentially expressed genes (MSE_DE and PCC_DE) by 0.06 and up to 0.15, respectively, compared with zero-shot scFoundation, and by 0.02–0.05 and up to 0.04, respectively, compared with GEARS ²⁹ (Supplementary Table 5). These improvements highlight the importance of efficient adaptation for this task. scPEFT also outperformed a linear model baseline ³⁰ by 0.02 to 0.09 in MSE_DE and 0.02 to 0.15 in PCC_DE. However, none of the deep learning models surpassed the linear model when evaluated across all genes (Supplementary Note 9). scPEFT with the scGPT backbone performed comparably with fine-tuned scGPT and GEARS, except on the Norman dataset with two-gene perturbations (Supplementary Table 6). When evaluating all genes, no significant differences were observed among these models. Given its lower memory and time requirements, scPEFT offers a practical and accessible solution for predicting perturbation responses in the most differentially expressed genes.

		Notes:
GEARS-HV-002 FORMAL VALIDATION	Single-cell technologies: current and near future DOI: 10.1007/s11427-024-2813-y Article page: https://doi.org/10.1007/s11427-024-2813-y Full text 1: https://link.springer.com/content/pdf/10.1007/s11427-024-2813-y.pdf Controller-registered target occurrence locator(s): occ-0001 (document_ae8274926b66104044e8.txt:p-0004)	Review status: Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: One of the most impressive applications of single-cell multi-omic technologies is the joint profiling of mRNA expression and epigenomic states. For example Notes:
GEARS-HV-006 FORMAL VALIDATION	Disentanglement of single-cell data with biolord DOI: 10.1038/s41587-023-02079-x Article page: https://doi.org/10.1038/s41587-023-02079-x Full text 1: https://www.nature.com/articles/s41587-023-02079-x.pdf Controller-registered target occurrence locator(s): occ-0001 (document_1075fecb1136e9480c16.txt:p-0003); occ-0002 (document_1075fecb1136e9480c16.txt:p-0003); occ-0003 (document_1075fecb1136e9480c16.txt:p-0003); occ-0004 (document_1075fecb1136e9480c16.txt:p-0003); occ-0005 (document_1075fecb1136e9480c16.txt:p-0003); occ-0006 (document_1075fecb1136e9480c16.txt:p-0003); occ-0007 (document_1075fecb1136e9480c16.txt:p-0005); occ-0008 (document_1075fecb1136e9480c16.txt:p-0009); occ-0009 (document_1075fecb1136e9480c16.txt:p-0009); occ-0010 (document_1075fecb1136e9480c16.txt:p-0009); occ-0011 (document_1075fecb1136e9480c16.txt:p-0009); occ-0012 (document_1075fecb1136e9480c16.txt:p-0009); occ-0013 (document_1075fecb1136e9480c16.txt:p-0009); occ-0014 (document_1075fecb1136e9480c16.txt:p-0009); occ-0015 (document_1075fecb1136e9480c16.txt:p-0012); occ-0016 (document_1075fecb1136e9480c16.txt:p-0012); occ-0017 (document_1075fecb1136e9480c16.txt:p-0012)	Review status: Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We show that biolord outperforms GEARS in the prediction of unseen one-gene perturbation (normalized mean squared error, one of one gene unseen; GEARS: 0.47; biolord: 0.37) and two-gene perturbations (normalized mean squared error, two of two genes unseen; GEARS: 0.53; biolord: 0.50, one of one gene unseen; GEARS: 0.39; biolord: 0.35, zero of two genes unseen; GEARS: 0.28; biolord: 0.20; Methods ; Supplementary Note 3 and Extended Data Fig. 2). Notes: